

Denoising Diffusion Probabilistic Models

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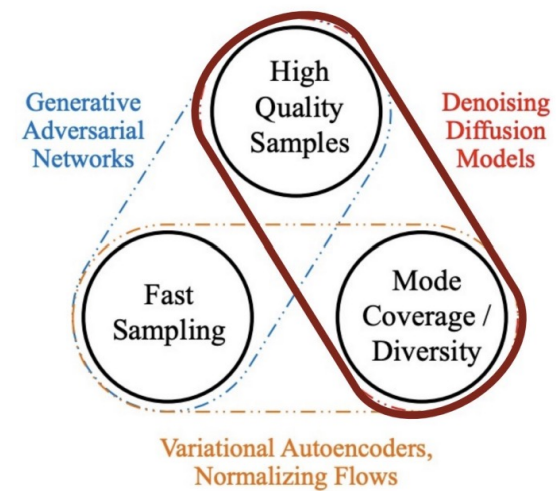
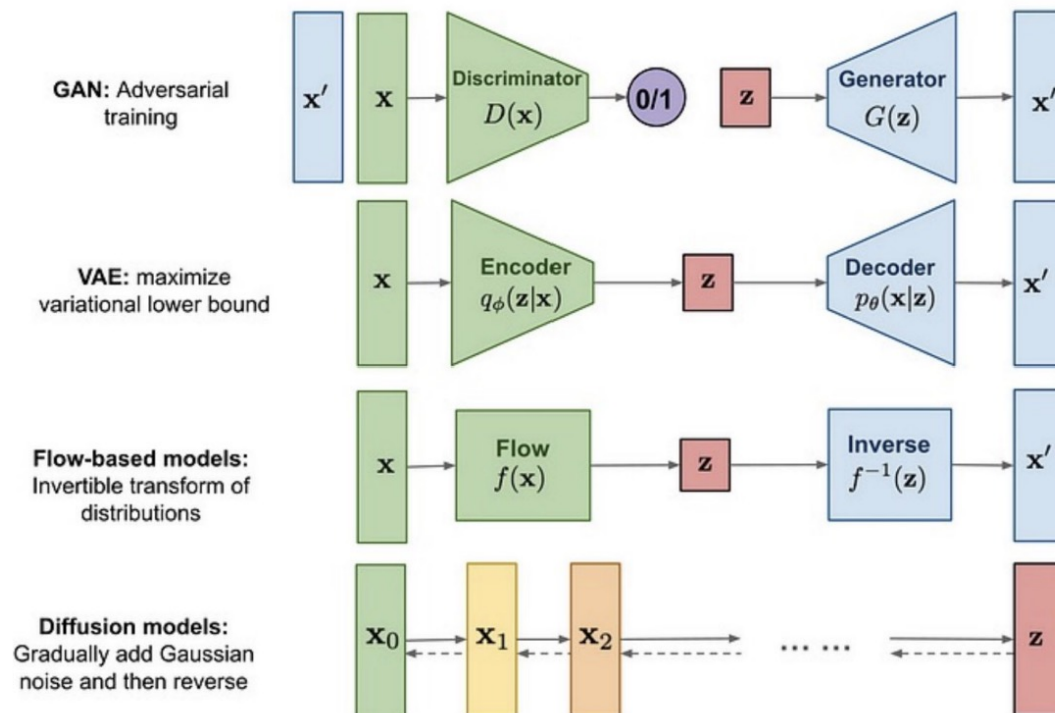
202455424 박현아

Paper Review

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Abstract & Introduction

03

- Diffusion probabilistic models은 nonequilibrium thermodynamics에서 영감을 받은 latent variable models이다.
- 모델은 Gaussian noise를 점진적으로 추가하는 forward diffusion process와 이를 제거하는 reverse process 으로 구성된다.
- DDPM은 각 단계에서 denoising score matching를 사용하여 추가되는 노이즈를 예측하며 신경망을 학습시킨다.

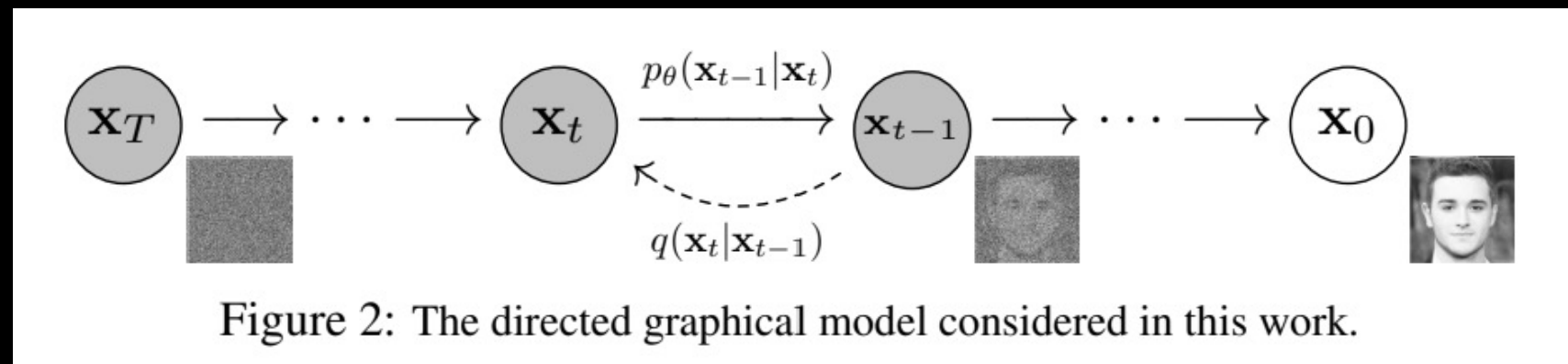
핵심 아이디어

$q(x_{1:T} | x_0)$: fixed forward process $\rightarrow$ $p_\theta(x_{0:T})$: learned reverse process

Diffusion Probabilistic Model

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생성모델은 diffusion process를 역행하도록 학습된 매개변수 Markov chain이다!!



Forward procee

$$x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$$

Reverse process

$$x_T \rightarrow \dots \rightarrow x_1 \rightarrow x_0$$

Background

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Reverse process

$$p_{\theta}(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t), \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

Forward process

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

$\beta_t \in (0,1)$ 는 시간 t 에서 추가되는 Gaussian noise의 크기를 조절한다.

Forward Process

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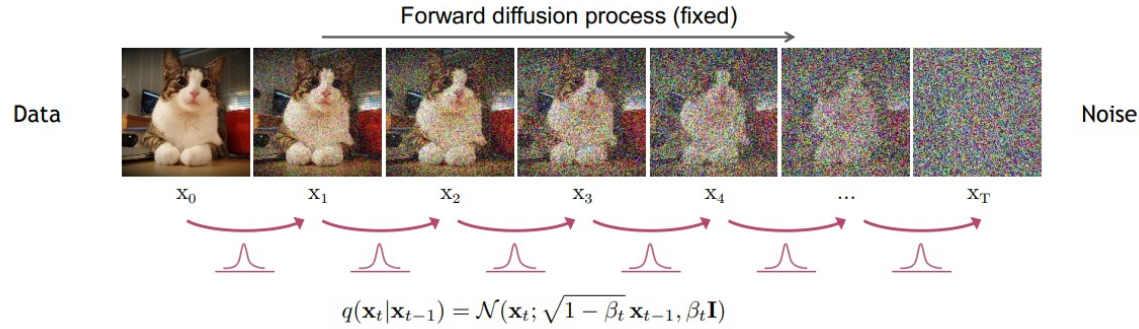
forward process q 는 고정되어 있으며, x_0 에 Gaussian noise를 점진적으로 추가한다.

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{(1-\beta_t)} x_{t-1}, \beta_t I)$$

$\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ 로 정의하면, x_t 는 x_0 에서 직접 샘플링할 수 있다.

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t)I)$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{(1-\bar{\alpha}_t)} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$



$$\alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \quad \text{Reparameterization Trick} \quad \epsilon_{t-1}, \epsilon_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon_{t-2}) + \sqrt{1 - \alpha_t} \epsilon_{t-1} \quad \text{윗줄 정의에 따라 } \mathbf{x}_{t-1} \text{을 재귀적으로 대입}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \epsilon_{t-2} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \quad \text{괄호 풀기}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \tilde{\epsilon}_{t-2} \quad \text{두 개의 가우시안 분포는 하나의 가우시안 분포로 합칠 수 있음}$$

$$= \dots \quad \mathcal{N}(\mathbf{0}, \sigma_1^2 \mathbf{I}) + \mathcal{N}(\mathbf{0}, \sigma_2^2 \mathbf{I}) = \mathcal{N}(\mathbf{0}, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$$

$$= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad \mathbf{x}_0 \text{까지 반복해서 대입 \& 정리}$$

$$\therefore q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad \text{Reparameterization Trick}$$

Reverse Process

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reverse process는 평균과 분산이 θ 로 매개변수화된 Gaussian transition으로 모델링된다.

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} ; \mu_{\theta}(\mathbf{x}_t, t), \Sigma_{\theta}(\mathbf{x}_t, t))$$

$\mathbf{x}_0$ 가 주어졌을 때의 순방향 posterior:

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} ; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t I)$$

학습은 실제 posterior mean $\tilde{\mu}_t$ 와 모델 평균 μ_{θ} 사이의 차이를 줄이는 방향으로 진행된다.

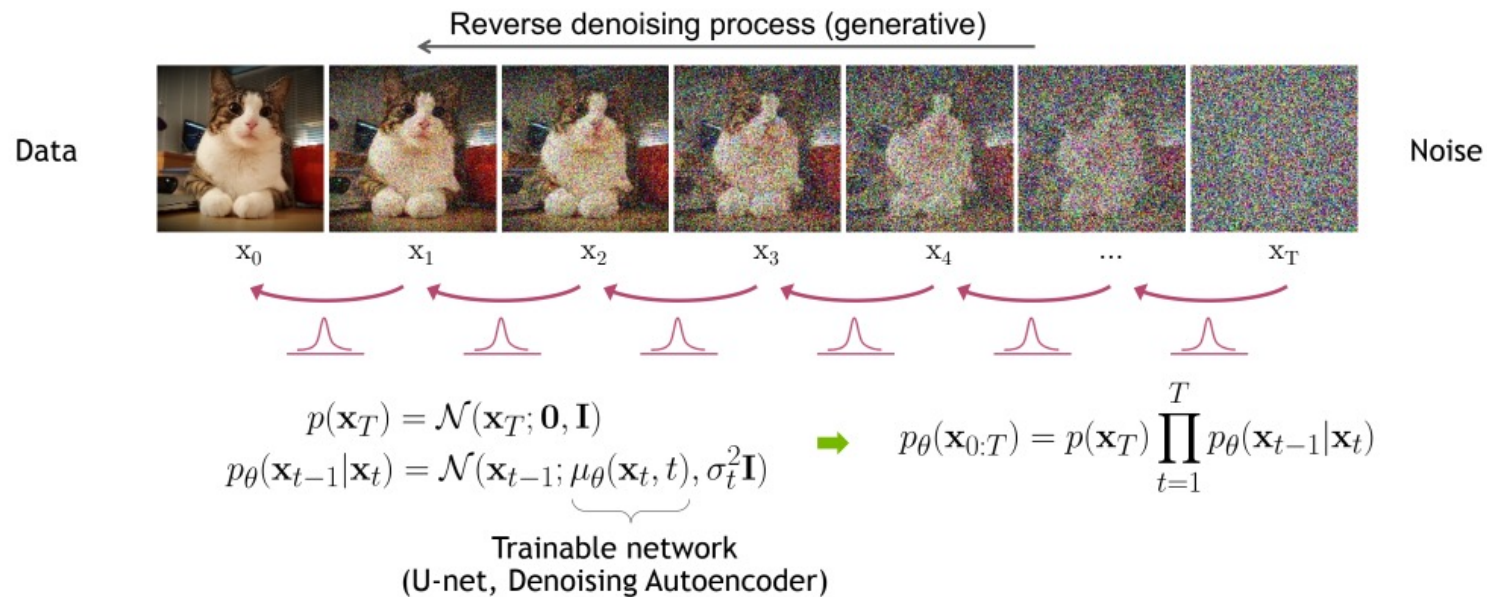
$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta}(\mathbf{x}_t, t)\|^2 \right] + C$$

$$q(X_{1:T} | X_0) = \prod_{t=1}^T q(X_t | X_{t-1})$$

사건 $A_1, A_2, \dots, A_n$ 이 발생가능하면,

$$\begin{aligned} &P(A_1 \cap A_2 \cap \dots \cap A_n) \\ &= P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 \cap A_2) \cdot \dots \cdot P(A_n | A_1 \cap A_2 \cap \dots \cap A_{n-1}) \end{aligned}$$

이 성립한다.



Training Objective

DDPM은 negative log-likelihood의 variational bound를 최소화하도록 학습한다.

$$E[-\log p_{\theta}(x_0)] \leq L$$

$$L = E_q [-\log p_{\theta}(x_{0:T}) / q(x_{1:T} | x_0)]$$

이를 timestep별 항으로 분해하면 다음과 같다.

$$L = L_T + \sum_{t>1} L_{t-1} + L_0$$

- L_T : $q(x_T | x_0)$ 와 prior $p(x_T)$ 사이의 KL divergence
- L_{t-1} : $q(x_{t-1} | x_t, x_0)$ 와 $p_{\theta}(x_{t-1} | x_t)$ 사이의 KL divergence
- L_0 : x_0 의 negative log-likelihood

따라서 reverse process 학습은 각 timestep에서 posterior와 model transition을 맞추는 문제로 정리된다.

ε -Prediction

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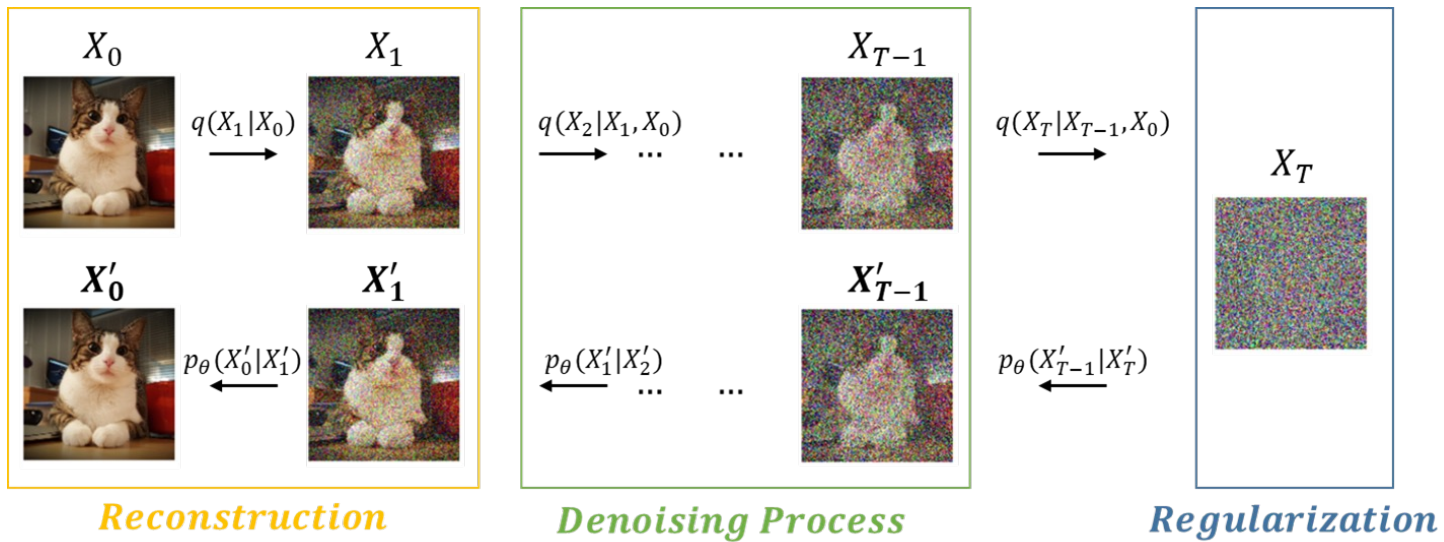
DDPM은 noise ε 를 예측하는 network $\varepsilon\theta(x_t, t)$ 를 이용해 역방향 과정의 평균을 매개변수화한다.

$$\mu\theta(x_t, t) = 1/\sqrt{\alpha_t} \cdot (x_t - \beta_t/\sqrt{(1-\bar{\alpha}_t)} \cdot \varepsilon\theta(x_t, t))$$

이 매개변수화를 사용하면 loss function은 다음과 같이 단순화 할 수 있다!

$$L_{\text{simple}}(\theta) = E_{t, x_0, \varepsilon} [\| \varepsilon - \varepsilon\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{(1-\bar{\alpha}_t)} \varepsilon, t) \|^2]$$

- 데이터 분포에서 x_0 를 샘플링한다.
- 시간 단계 t 와 Gaussian noise ε 를 샘플링한다.
- x_t 를 구성하고, $\varepsilon\theta$ 가 ε 를 예측하도록 MSE로 학습한다.



Negative Log Likelihood ($\mathbb{E}_{X_T \sim q(X_T | X_0)} [-\log p_\theta(X_0)]$)

$$= \mathbb{E}_q \left[\underbrace{D_{KL}(q(X_T|X_0) \parallel p(X_T))}_{\text{Regularization}} + \underbrace{\sum_{t=2}^T D_{KL}(q(X_{t-1}|X_t, X_0) \parallel p_\theta(X_{t-1} | X_t))}_{\text{Denoising Process}} + \underbrace{\log p_\theta(X_0|X_1)}_{\text{Reconstruction}} \right]$$

Log-likelihood 최대화

학습 데이터 $x_0 \sim q(x_0)$ 에서 시작하여 forward process를 통해 $q(x_t|x_0)$ 로 x_t 를 만든다.

그 상태에서 reverse 분포인 $p_\theta(x_{t-1}|x_t)$ 가 posterior $q(x_{t-1}|x_t, x_0)$ 를 잘 맞추도록 학습한다.

모델 파라미터 θ 는 p_θ 에 존재한다.

따라서 학습 목표는 데이터의 로그가능도 $\log p_\theta(x_0)$ 를 최대화 하는 것이다.

(또는, $-\log p_\theta(x_0)$ 를 최소화: Minimize Negative Log-likelihood)

$x_0 \sim q(x_0)$ 에 대하여

$$\max_{\theta} E_{q(x_0)}[\log p_\theta(x_0)]$$

적분하기 쉽지 않음;

$$\begin{aligned}
 \log p_{\theta}(x_0) &= \log \int p_{\theta}(x_{0:T}) dx_{1:T} \\
 &= \log \int q(x_{1:T}|x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} dx_{1:T} \quad \text{우리가 아는 값 도입!} \\
 &= \log E_{q(x_{1:T}|x_0)} \left[\frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right] \quad E_q[f(X)] = \int q(X)f(X)dX \\
 &\geq E_{q(x_{1:T}|x_0)} \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right] \quad (\because \text{Jensen's inequality}) \\
 &\quad \log E[Y] \geq E[\log Y]
 \end{aligned}$$

$$\text{ELBO} := E_{q(x_{1:T}|x_0)} [\log p_{\theta}(x_{0:T}) - \log q(x_{1:T}|x_0)]$$

$$\begin{aligned}
 q(x_{1:T}|x_0) &= \prod_{t=1}^T q(x_t|x_{t-1}) & \log q(x_{1:T}|x_0) &= \sum_{t=1}^T \log q(x_t|x_{t-1}) \\
 p_{\theta}(x_{0:T}) &= p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1}|x_t) & \log p_{\theta}(x_{0:T}) &= \log p(x_T) + \sum_{t=1}^T \log p_{\theta}(x_{t-1}|x_t)
 \end{aligned}$$

$$\text{ELBO} = E_q \left[\log p(x_T) + \sum_{t=1}^T \log p_{\theta}(x_{t-1}|x_t) - \sum_{t=1}^T \log q(x_t|x_{t-1}) \right]$$

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

$$\log q(x_t|x_{t-1}) = \log q(x_{t-1}|x_t, x_0) + \log q(x_t|x_0) - \log q(x_{t-1}|x_0)$$

$$\begin{aligned} \sum_{t=1}^T \log q(x_t|x_{t-1}) &= \log q(x_1|x_0) + \sum_{t=2}^T \log q(x_t|x_{t-1}) \\ &= \log q(x_1|x_0) + \sum_{t=2}^T [\log q(x_{t-1}|x_t, x_0) + \log q(x_t|x_0) - \log q(x_{t-1}|x_0)] \\ &= \sum_{t=2}^T \log q(x_{t-1}|x_t, x_0) + \log q(x_T|x_0) \quad (\text{telescoping}) \end{aligned}$$

이 식을 ELBO에 대입하고 $t = 1$ 항을 마지막에 따로 남겨두면

$$\begin{aligned} \text{ELBO} &= E_q[\log p(x_T) - \log q(x_T|x_0)] \\ &\quad + \sum_{t=2}^T E_q[\log p_\theta(x_{t-1}|x_t) - \log q(x_{t-1}|x_t, x_0)] \\ &\quad + E_q[\log p_\theta(x_0|x_1)] \end{aligned}$$

첫번째 항은 KL Divergence로 바꿔서 쓰면 아래와 같다.

$$(I) = E_q[\log p(x_T) - \log q(x_T|x_0)] = -\text{KL}(q(x_T|x_0)||p(x_T))$$

$$\text{KL}(P||Q) = \int p(x) \log \frac{p(x)}{q(x)} dx = E_{x \sim p(x)} \left[\log \frac{P(X)}{Q(X)} \right]$$

$$E_{q(x_{t-1}|x_t, x_0)} [\log p_\theta(x_{t-1}|x_t) - \log q(x_{t-1}|x_t, x_0)]$$

$$= -E_q \left[\log \frac{q(x_{t-1}|x_t, x_0)}{p_\theta(x_{t-1}|x_t)} \right].$$

$$\mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)}_{L_0} \right] \quad (5)$$

L_T : Forward Process에서 원본 이미지를 얼마나 가우시안 노이즈와 비슷하게 잘 바꾸냐?
 -> 베타 값이 상수이므로, 최적화와는 무관한 고정값

L_{t-1} : Reverse Process의 각 스텝에서 이전 스텝의 (노이즈를 살짝 덜어낸) 이미지를 얼마나 잘 복원하는지를 측정

L_0 : x1에서 x0을 복원하는 과정
 -> 실험적으로도 큰 영향을 주지 않음

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I) \quad p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, x_t), \sigma_t^2 I)$$

두 분포 모두 가우시안이고 분산을 $\sigma_t^2 = \tilde{\beta}_t^2$ 로 두면

$$\text{KL} = \frac{1}{2\sigma_t^2} \|\tilde{\mu}_t - \mu_\theta\|^2 + \text{const.}$$

$$\hat{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta)$$

$$\mu_\theta(\mathbf{x}_t, t) = \tilde{\mu}_t\left(\mathbf{x}_t, \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(\mathbf{x}_t))\right) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t)\right) \quad (11)$$

$$\begin{aligned} & q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \\ &= q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \quad \text{t-1에 대한 분포를 t에 대한 분포로 변형} \\ &= \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{q(\mathbf{x}_t|\mathbf{x}_0)} \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \quad \text{마르코프 성질 적용 (x0 제거)} \\ &\propto \exp\left(-\frac{(\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{2\beta_t}\right) \frac{\exp\left(-\frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{2(1-\bar{\alpha}_{t-1})}\right)}{\exp\left(-\frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{2(1-\bar{\alpha}_t)}\right)} \quad \begin{array}{l} \text{가우시안 분포의 확률밀도함수(PDF) 수식 대입} \\ \text{'='이 아닌 ':'인 이유: exp의 계수를 생략하여 표기} \end{array} \\ &\quad \text{Forward Process의 마르코프 체인 정의} \quad \begin{array}{l} q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}) \\ q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \end{array} \\ &\quad \text{가우시안 분포 PDF 수식} \quad \mathcal{N}(x, \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \\ &= \exp\left(-\frac{(\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{2\beta_t}\right) + \frac{-(\mathbf{x}_t - \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{2(1 - \bar{\alpha}_{t-1})} - \frac{-(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{2(1 - \bar{\alpha}_t)} \quad \begin{array}{l} \text{지수 법칙 적용} \\ \text{(일미 같으면 곱 → 합, 나누기 → 차)} \end{array} \\ &\quad \text{제곱 항 전개} \\ &= \exp\left(-\frac{(\mathbf{x}_t^2 - 2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1} + \alpha_t\mathbf{x}_{t-1}^2)}{2\beta_t}\right) + \frac{-(\mathbf{x}_t^2 - 2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1} + \bar{\alpha}_{t-1}\mathbf{x}_0^2)}{2(1 - \bar{\alpha}_{t-1})} - \frac{-(\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{2(1 - \bar{\alpha}_t)} \\ &= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}}\right)\mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}}\mathbf{x}_0\right)\mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0)\right)\right) \quad \begin{array}{l} \mathbf{x}_{t-1}\text{항에 대하여 정리} \\ \text{그 외 항은 상수로 취급} \end{array} \\ &= \exp\left(-\frac{1}{2}\left(\left(\frac{1 - \bar{\alpha}_t}{\beta_t(1 - \bar{\alpha}_{t-1})}\right)\mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t}\mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}}\mathbf{x}_0\right)\mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0)\right)\right) \quad \mathbf{x}_{t-1}\text{ 계수 정리} \\ &= \exp\left(-\frac{1 - \bar{\alpha}_t}{2\beta_t(1 - \bar{\alpha}_{t-1})}(\mathbf{x}_{t-1}^2 - 2\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0\right)\mathbf{x}_{t-1}) + C'(\mathbf{x}_t, \mathbf{x}_0)\right) \\ &\quad \begin{array}{l} \mathbf{x}_{t-1}\text{'의 계수가 1이 되도록 묶고} \\ \text{가우시안 분포 PDF의 exp의 지수 항과 비교} \end{array} \quad \left(-\frac{(x - \mu)^2}{2\sigma^2}\right) = \left(-\frac{x^2 - 2\mu x + \mu^2}{2\sigma^2}\right) \end{aligned}$$

따라서, 만약 $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I})$ 으로 둔다면,

$$\begin{aligned} \bullet \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 \\ \bullet \tilde{\beta}_t &= \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t \end{aligned}$$

$$\begin{aligned}\tilde{\mu}_t(x_t, t) - \mu_\theta(x_t, t) &= \left(\frac{1}{\sqrt{\alpha_t}} x_t - \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon \right) - \left(\frac{1}{\sqrt{\alpha_t}} x_t - \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) \\ &= \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} (\epsilon_\theta(x_t, t) - \epsilon)\end{aligned}$$

$$D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))$$

$$= \frac{1}{2\sigma_t^2} \|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2 \quad \text{가우시안 분포의 KLD}$$

$$= \frac{1}{2\sigma_t^2} \left\| \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) - \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) \right\|^2 \quad \text{평균 대입}$$

$$= \frac{1}{2\sigma_t^2} \left\| \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} (\epsilon - \epsilon_\theta(\mathbf{x}_t, t)) \right\|^2 \quad \text{괄호 풀고 정리}$$

$$= \frac{(1 - \alpha_t)^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|(\epsilon - \epsilon_\theta(\mathbf{x}_t, t))\|^2 \quad \text{상수 밖으로 빼기}$$

$$= \frac{(1 - \alpha_t)^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|(\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t))\|^2 \quad \mathbf{x}_t \text{ 대입}$$

$$\therefore L_{t-1} = \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{(1 - \alpha_t)^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|(\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t))\|^2 \right]$$

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

Training & Sampling

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Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t)\|^2$   
6: until converged
```

Training

t 를 무작위로 선택하고 $\mathbf{x}_t$ 를 합성한 뒤, noise-prediction loss로 θ 를 업데이트한다.

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 에서 시작하여 $\mathbf{x}_{t-1}$ 을 반복적으로 샘플링하고 최종적으로 $\mathbf{x}_0$ 를 얻는다.

Model Architecture & Implementation

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- Backbone: Wide ResNet 기반 U-Net architecture
- Timestep t 는 sinusoidal position embedding으로 입력된다.
- 16×16 feature-map resolution에서 self-attention을 사용한다.
- Diffusion length: $T = 1000$.
- forward process의 variance $\beta_1, \dots, \beta_T$ 는 10^{-4} 에서 0.02까지 선형적으로 증가하는 고정 상수이다
- 실험에서는 reverse process의 분산을 학습하지 않고, t 에 따라 미리 정해진 상수로 설정한다.

Network output은 x_t 와 같은 차원을 가지며 $\epsilon\theta(x_t, t)$ 를 나타낸다.

Experiments

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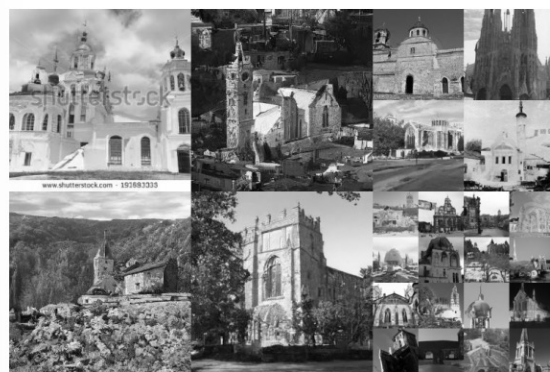


Figure 3: LSUN Church samples. FID=7.89



Figure 4: LSUN Bedroom samples. FID=4.90

Algorithm 3 Sending $\mathbf{x}_0$

- 1: Send $\mathbf{x}_T \sim q(\mathbf{x}_T | \mathbf{x}_0)$ using $p(\mathbf{x}_T)$
- 2: **for** $t = T - 1, \dots, 2, 1$ **do**
- 3: Send $\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0)$ using $p_\theta(\mathbf{x}_t | \mathbf{x}_{t+1})$
- 4: **end for**
- 5: Send $\mathbf{x}_0$ using $p_\theta(\mathbf{x}_0 | \mathbf{x}_1)$

Algorithm 4 Receiving

- 1: Receive $\mathbf{x}_T$ using $p(\mathbf{x}_T)$
- 2: **for** $t = T - 1, \dots, 1, 0$ **do**
- 3: Receive $\mathbf{x}_t$ using $p_\theta(\mathbf{x}_t | \mathbf{x}_{t+1})$
- 4: **end for**
- 5: **return** $\mathbf{x}_0$

CIFAR-10

Inception Score

9.46

FID

3.17

LSUN 256×256

ProgressiveGAN과 비교 가능한 sample quality를 보였다.

Conclusion & Summary

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- DDPM은 점진적인 Gaussian diffusion의 forward 과정과 reverse의 이미지 생성 과정으로 나뉜다.
- 간단한 noise prediction objective만으로도 reverse process를 효과적으로 학습할 수 있다.
- Adversarial training 없이도 고품질의 이미지를 생성할 수 있다.
- Sampling 과정에서는 많은 timestep에 걸쳐 reverse Markov chain을 수행해야 하므로, 한 번의 forward pass로 생성하는 모델에 비해 계산 비용이 크다.
- 논문은 diffusion model과 denoising score matching의 관계를 제시하고, 이를 progressive lossy decomposition의 관점에서도 해석한다.